Degrees of freedom.

Grubler's / Kutzbach Criteria.

$$DOF = 3(n-1) - 2j - h - F_r$$

n = no of links

j = no of joints (Binary)

h = no of higher Pairs

f_r = Redundant DOF

Teinery Joint = $j+1$

<u>links</u>

→ fixed links

→ Binary

→ Terinery

→ Quartineey links.

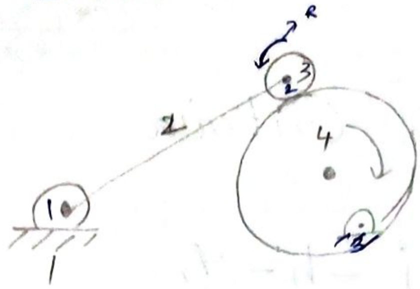
if DOF = 0 structure

DOF = −1 Super structure

DOF => 1 Mechanism.

Degrees of freedom.

(1)



$n = 4$

$j = 3$

$h = 1$

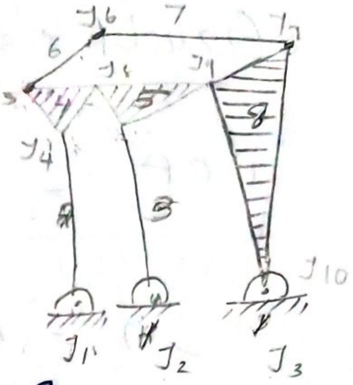
$F_r = 1$

$3(n-1) - 2j - h - F_r$

$3(3) - 2(3) - 1 - 1$

$n = 1$

(2)



$n = 8$

$j = 10$

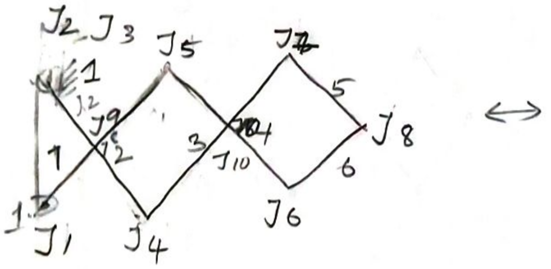
$h = 0$

$F_r = 0$

$3(n-1) - 2j - h - F_r$

$= 3(7) - 2(10)$ $= 1$

(3)



$n = 8$

$J = 10$

$h = 0$

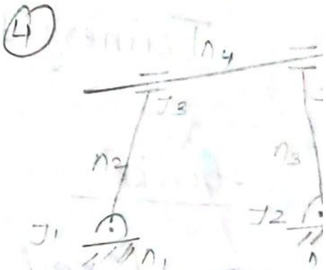
$f_r = 0$

$3(n-1) - 2j - h - f_r$

$3(7) - 2 \times 10 - 0 - 0$

$= 1$

(4)



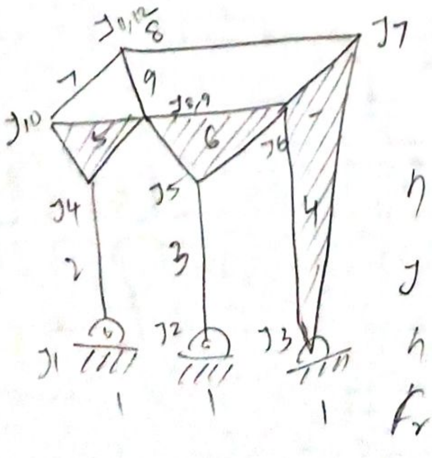
$J = 4$

$n = 4$

$F_r = 1$

$3(4-1) - 2 \times 4 - 1$

$= 0$



$n = 9$

$J = 12$

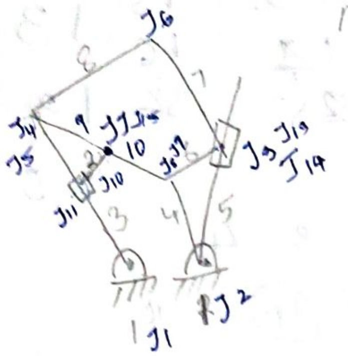
$h = 0$

$F_r = 0$

its a structure

$3(8) - 2 \times 12 - 0 - 6$

$= 24 - 24 = 0$

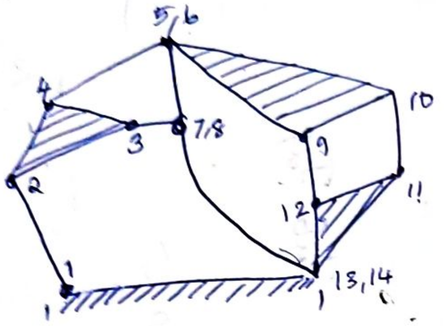


$n = 12$

$J = 15$

$3(11) - 15\times 2 - 0 - 0$

$= 3$



$3(n-1) - 2j - h - F_r$.

$n = 11$

$j = 14$

$3(10) - 2\times 14$

$= 30 - 28 = 2$

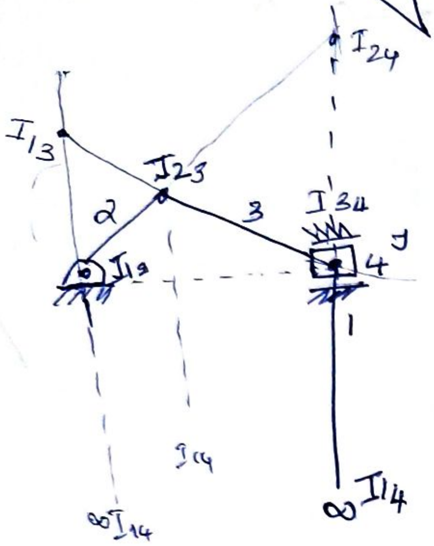
Velocity Analysis

- Graphical method
  - ICR
  - Relative Velocity method
- Analytical Method

Instantaneous Centre of Rotation

Aronhold Instantaneous

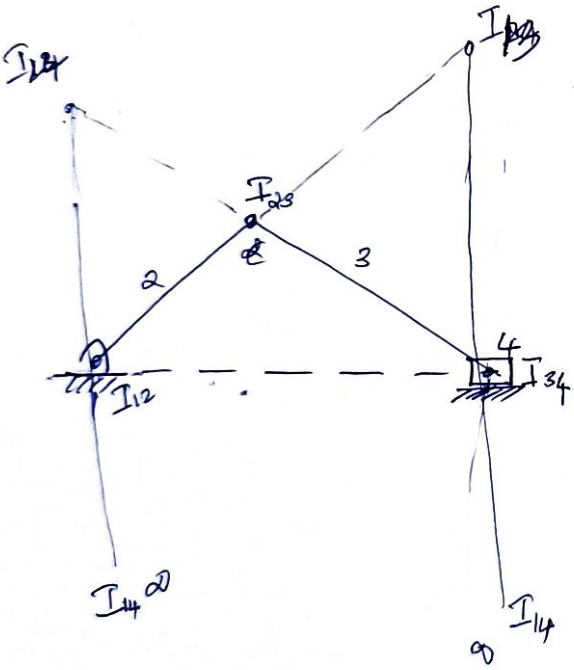
No: of ICR $= \dfrac{n(n-1)}{2}$, n = no: of links

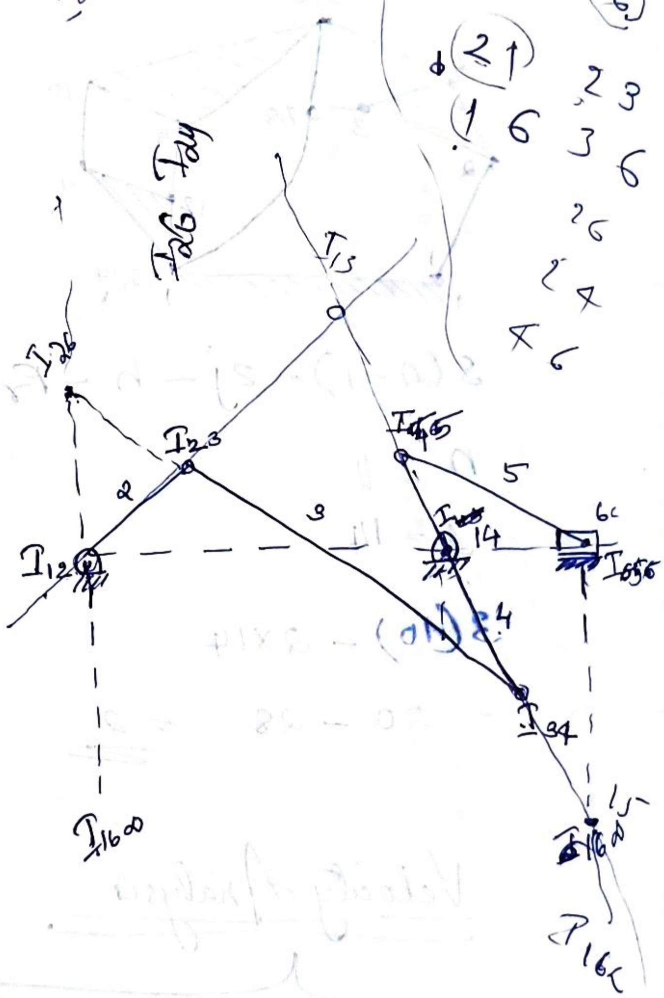


15 15 13 13
12 16
25 56 12 14
23 43

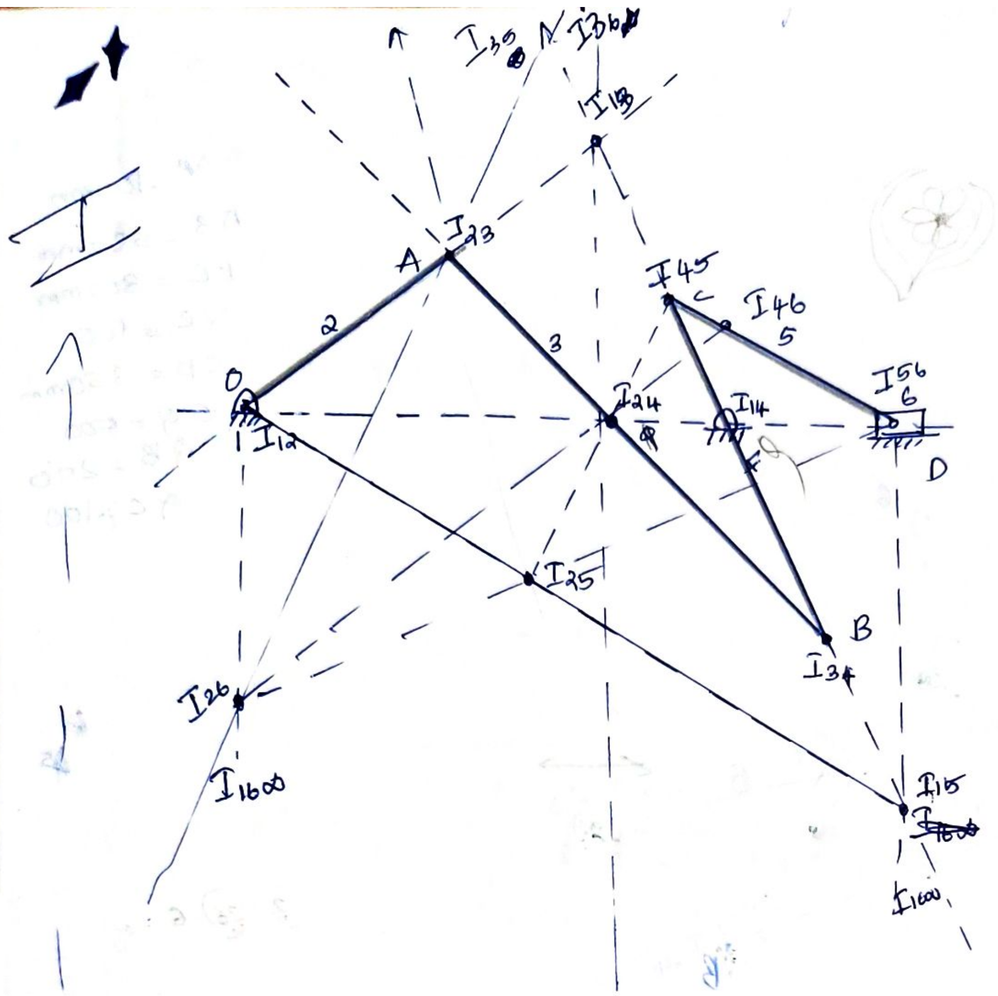
1 2 3 4
12 23 34
13 24
14

26 26
21 23
16 36
26
24
46





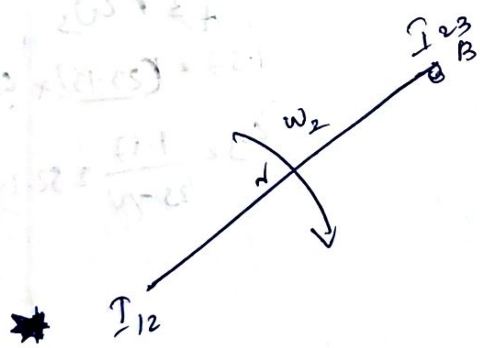
1 2 3 4 5 6
12 23 34 45 56
13 24 35 46
14 25 36
15 26
16



1 2 3 4 5 6

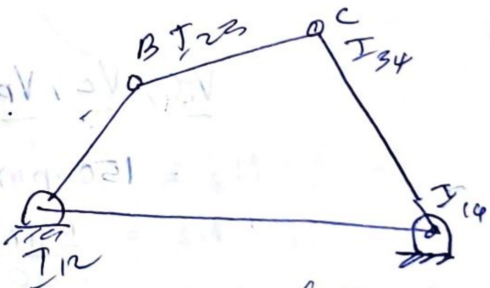
12 23 34 45 56
13 24 35 46
14 25 36
15 26
16

Next find 13 as go coloumn vise so find 2 pairs with a common number. Having 1 & 3 they are 12, 23 and 14, 34



$V = r\omega$

$V_{23} = V_B = (23-12) \times W_2$



find velocity at C

The dimensions of the link OA = 100mm
refer fig I

AB = 580mm
BC = 300mm
QC = 100 mm
CD = 350 mm

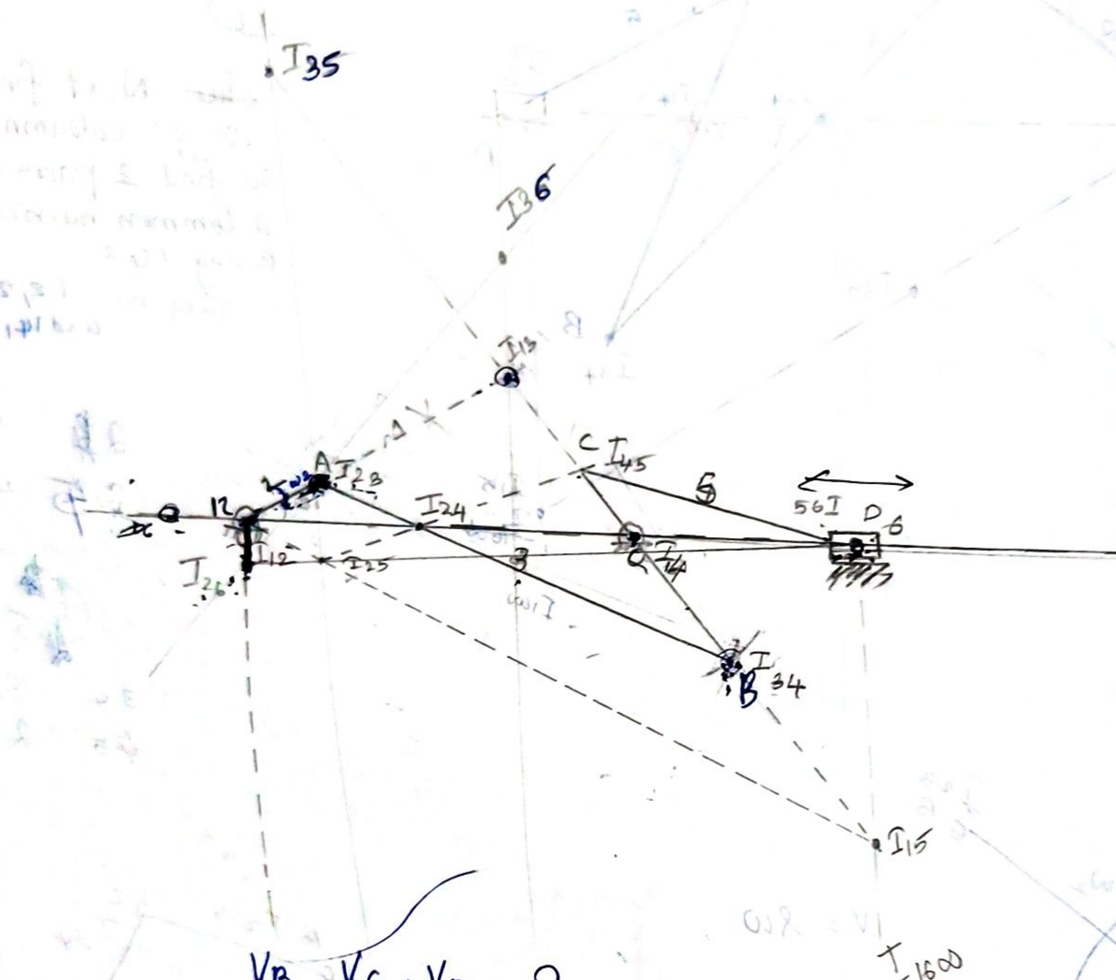
the crank OA rotates at 150 rpm. for the position when the crank OA makes an angle of 80° with the horizontal

(1) Find linear velocity of the pivot points B, C, D
(2) Find Angular velocity of the links AB BC & CD

Scale 1:10

Δ

OA = 100 mm
AB = 580 mm
BC = 300 mm
QC = 100
CD = 350 mm
OQ = 500
QB = 200
QC = 100



2 26 6

$V_A = r\omega_3$

$1.57 = (23-13) \times \omega_3$

$\omega_3 = \frac{1.57}{(23-13)} = 52.33$

$\underline{V_B}, \underline{V_C}, \underline{V_D} = ?$

$N_2 = 150$ rpm.

$\omega_2 = \frac{2\pi N}{60} = \frac{2 \times \pi \times 150}{\cancel{60}\ 8} = \cancel{0.0368}.5\pi = \cancel{0.2617}\ \underline{15.707}$

$V_{23} = r\omega_2 = (I_{23} - I_{12}) \times \omega_2 = 0.1 \times 15.7 = \underline{1.57\ m/s}$

Also

$V_{23} = \omega_3(I_{13} - I_{23})$

$\omega_3 = \frac{1.57}{0.03} = \underline{52.33}$

ω_3 at both ends of a link is same.

$V_B = V_{34} = r\omega_3 = 0.045 \times 52.33 = \underline{2.354\ m/s}$

$(13-34) \times 52.33 =$

$\cancel{V_B = V_{45} = \omega_4}$

$V_B = \omega_4 (14-34)$

$\omega_4 = \underline{\underline{\quad}}$.

$V_C = \omega_4 (14.45) = 1.2 m/s$.

$V_C = V_{45} = \omega_5 \times (15.45)$

$\omega_5 = \dfrac{V_{45}}{(15-45)} =$

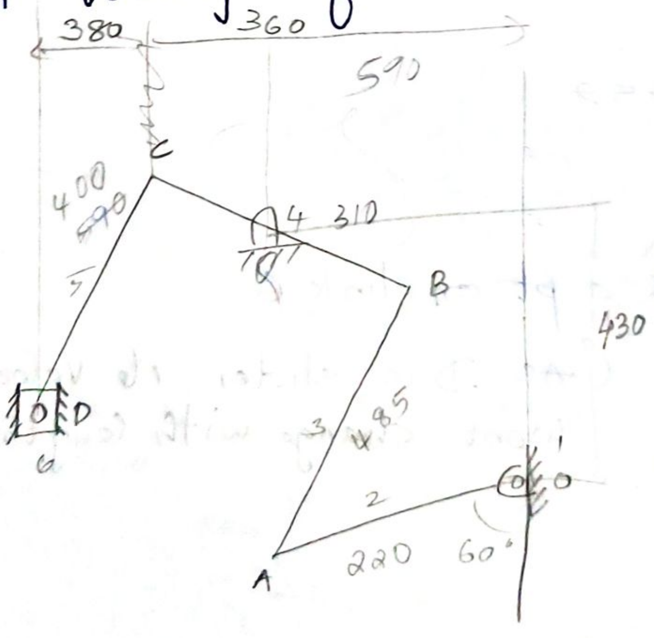
$V_{66} = \omega_5 \times 15 =$

$V_D = \dfrac{V_{45} \times (15.56)}{15-45} =$

$\underline{\underline{V_D = 0.788}}$

1 cm = 0.01 = 10 mm

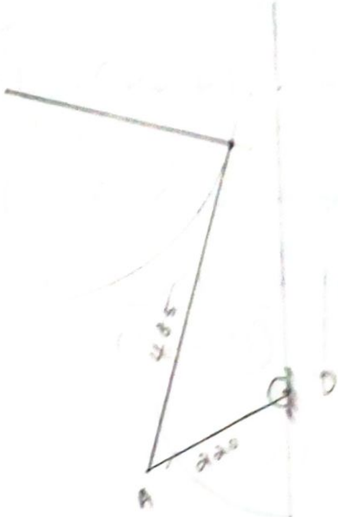
(9) Find Velocity of slider D.



for a slider joint velocity.

(i) Only have to find ~~the~~ one IC connecting directly to link D.

(ii) Asume a point in D as a point on a link whose ω is given here link 2 so find I_{26}.



Assuming 26 as a point on link 2

$\omega_2 = 5\pi \left(\frac{2\pi \times 150}{60}\right)$

$= 1.57 \text{ rad/s}$

$V_2 = r\omega_2$

$= (12-26) \times 1.57$

$= 2.28 \text{ m/s}$

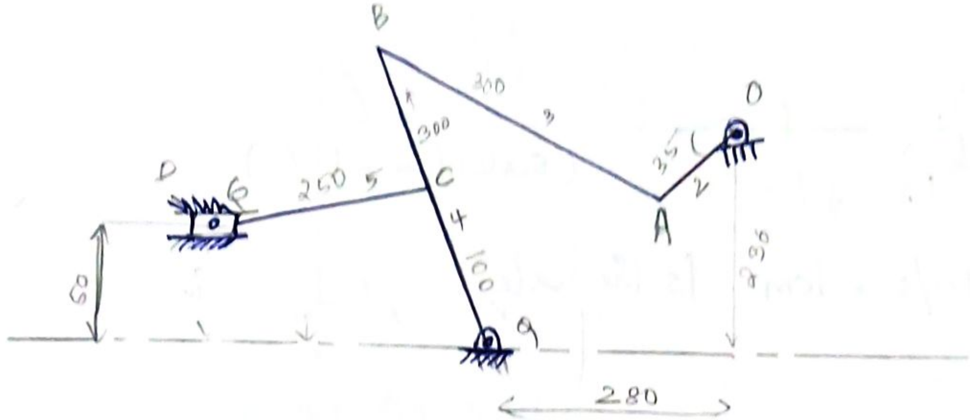
Considering 26 as a pt on link 6.

$V_D = 2.28$

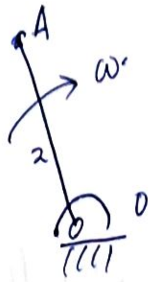
(As D is slider its velocity wont change with length (radius)

Relative Velocity method.

Scale 1:5



Configuration diagram → Drawn with scale
or Displacement Diagram



$\vec{V_A} = \vec{V_O} + \vec{V_{AO}}$ — Vectorial addition

Velocity A = Velocity at O + Velocity of A relative to O

- line of action of a Relative Velocity will always be Perpendicular to it.

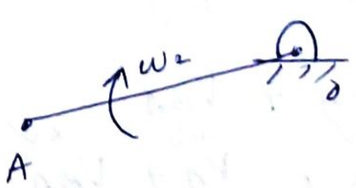
Velocity Diagram

$\omega_2 = \frac{2\pi N_2}{60} = 5\pi$

$\vec{V_A} = \vec{V_O} + \vec{V_{AO}}$
(Absolute) (Absolute) (relative)

- Now we draw this equation
- Velocity Polygon.
- fixed pole represent the Velocity of fixed points in our question → O

link 2



$\vec{V_A} = \cancel{\vec{V_O}} + \vec{V_{AO}}$

$\rightarrow V_A = V_{AO} = r_{OA} \times \omega_2 = 0.1 \times 5\pi = 1.57\ m/s$

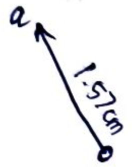
for Direction
Relative Vector is $\perp$ to OA and in Direction of rotation.
All absolute values shoud.
• Always be drawn away from fixed pole. (Absolute values ...

To Draw

1.57 m/s $\Rightarrow$ 1.5 cm.

1 ~~cm =~~ (Scale 1cm = 1m/s)

$\Rightarrow$ 1m/s = 1cm is the scale



fixed pole.

Now Considering Point A as a point on both link 3 and 2,

Considering it as on 3

$$\vec{V_B} = V_A + \overleftrightarrow{V_{BA}}$$

here we dont know D and M of V_B

Similarly Magn of V_{BA} is also unknown hence.

Velocity Diagram Canot be drawn with 3 unknown

So Consider B as a point on link 4

$$V_B = V_Q + V_{BQ}$$

here V_Q is fixed hence $V_Q = 0$.

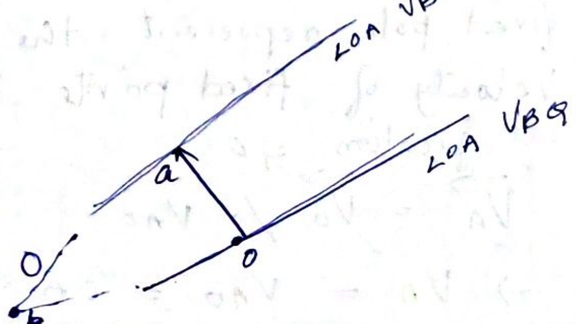
$$V_B = V_{BQ}$$ (we know LOA of V_{BQ}

$$V_B = V_A + V_{BA}$$

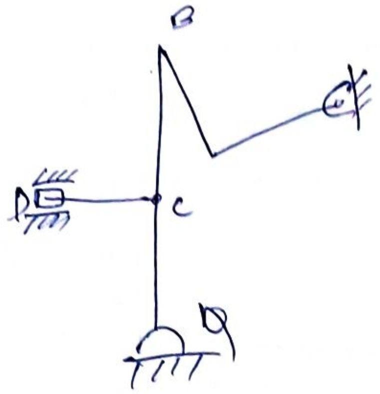
$$\Rightarrow V_{BQ} = V_A + V_{BA}$$

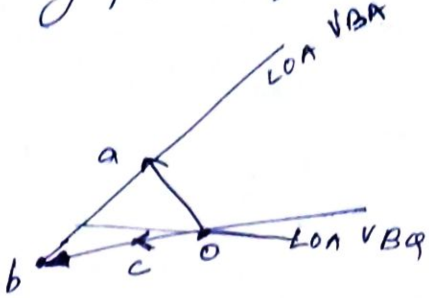
LOA VBQ from a as Q is fixed

... Point from fixed pole o



Since B is a point on the line of action of V_{BA} and V_{B} the intersection give the point B. hence its magnitude for direction (All absolute vectors away from o.





By Proportion we can write

$$\boxed{\frac{QC}{QB} = \frac{OC}{OB}}$$

Since the link and velocity Diagrams are proportional.

$$\Rightarrow OC = \frac{QC}{QB} \times OB$$

$$= \frac{100}{30} \times OB = 0.3 \times OB$$

hence we got D & M of C

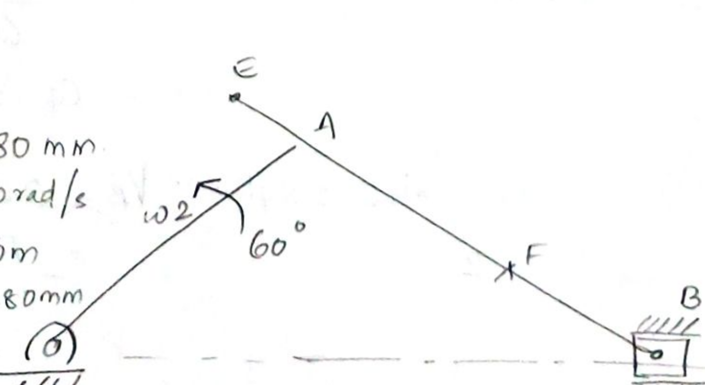
Now link 5 (slider)

is a point on 5

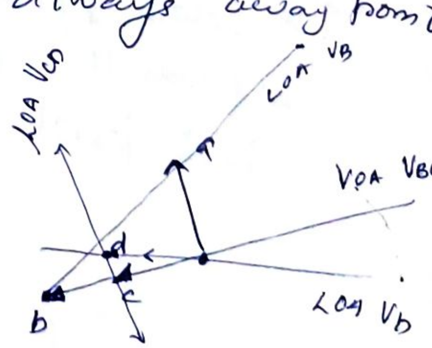
$$\overset{\times\checkmark}{V_D} = \overset{\checkmark\checkmark}{V_C} \rightarrow \overset{\times\checkmark}{V_{DC}}$$

Draw all absolute vectors from O and always away from O

(3)



OA = 480 mm
W_2 = 20 rad/s
EB = 1.6m
EA = 480mm



find

velocity of slider.

Velocity of E

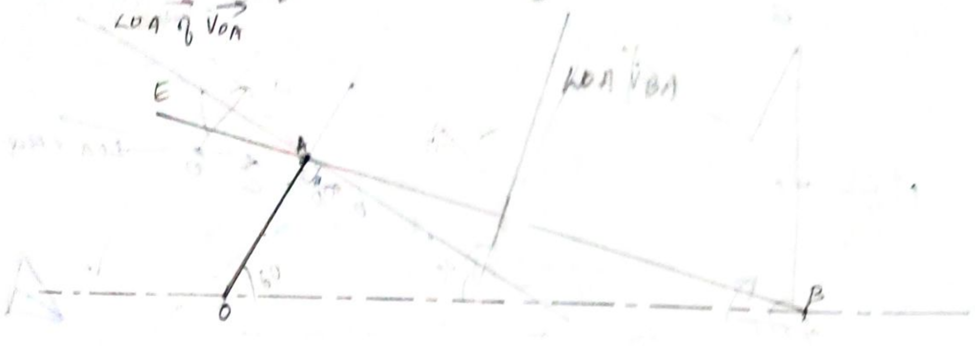
find position & velocity of E on the connecting rod such that absolute velocity of F is minimum.

V_{xy} → Relative

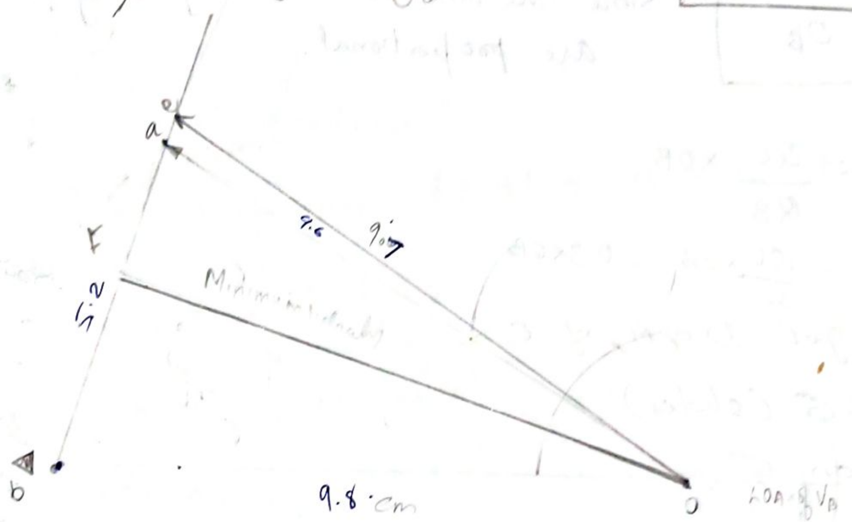
Scale 1:20

Configuration Diagram

OA = 480 mm
EB = 1600 mm
EA = 480 mm



Velocity Polygon Scale 1cm = 1m/s



$\vec{V_A} = \vec{V_O} + V_{AO}$

$\rightarrow \vec{V_A} = \vec{V_{AO}}$

Mag $\vec{V_{AO}} = r\omega$

$= 0.48 \times 20$

$= \cancel{4\,m/s}$

$9.6\,m/s$

Direction $\perp \vec{V_{AO}}$

$ea = \frac{EA}{BA} \times ba$

$= \frac{480}{1600} \times 5.2$

$ea = 0.415\,m/s$

The Velocity of point of F on the Connecting Rod such that absolute Velocity of F is minimum ~~is the point~~ the minimum distance will be $\perp$

$V_F = 9.4\,m/s$

$\frac{FB}{AB} = \frac{fb}{ba} \quad \frac{2.76}{5.2} \times 1600 = 840\,mm$

$V_B = OB \times scale$

$= 9.8\,m/s$

$V_E = OE \times scale$

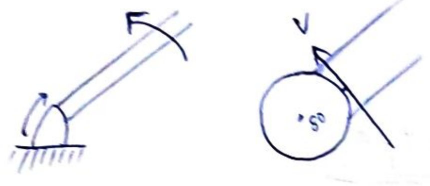
$= 9.7\,m/s$

~~r_{AB}~~ $V_{ab} = AB \times \omega_{ab}$

$\omega_{ba} = \dfrac{V_{ba}}{AB} = \dfrac{5.2}{1.5} = 3.25 m/s.$

1) find Velocity of rubbing ~~of the points~~ at the pins of crank-shaft, crank, connecting rod, crosshead having diameter, 80mm 60mm, 100mm respectively.

Velocity of rubbing



$V = \omega_2 \times radius$

$= 20 \times \dfrac{80}{2} \times \dfrac{1}{100} m. = 0.8 m/s$

At joints with both links having motion like in A check the direction of rotation; we have 2ω, ω_{ao} & ω_{ba}

if both rotates in same direction take difference

if Different Direction Sum it.

$V = r\omega$

Anti clock

$V = 0.03 \times (\omega_{ao} + \omega_{ba})$

As it is in opposite direction

$= 0.03 \times (20 + 3.28) = \underline{0.69} m/s$

Point B. $V = r\omega$

$= 0.05(3.28) = \underline{0.164 m/s}$

find the Velocity of slider (cutting tool)

P - Point Connecting link 2 & slider

Q → Point Connecting slider & 3

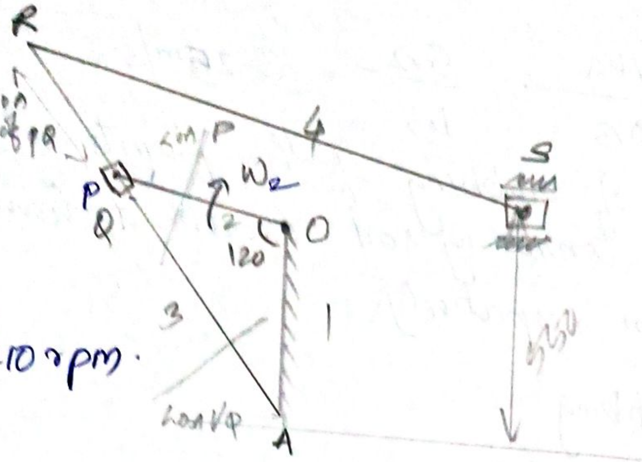
Given $OA = 400mm$

$OP = 200mm$

$AR = 700mm$

$RS = 300mm$

OP rotates @ 210 rpm.



$$\omega = \frac{2\pi N}{60}$$

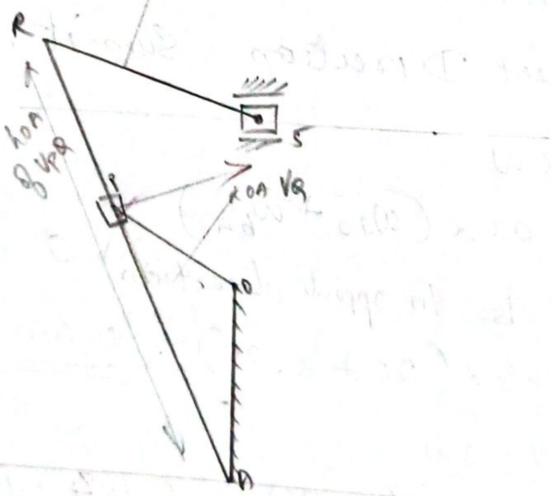
$$\omega_2 = \frac{2\pi \times 210}{60} = \underline{\underline{7\pi}} = \underline{\underline{22\ rad/s}}$$

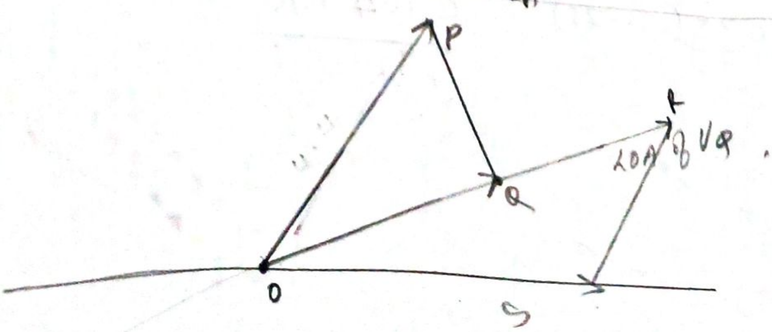
P in link 2

$$V_P = r\omega = 200 \times 22 = \underline{\underline{4.4\ m/s}}$$

$$\vec{V_Q} = \vec{V_P} + V_{PQ}$$

$\boxed{1:10}$





$V_Q = V_A + V_{AQ}$

to find Velocity of R

$\frac{RA}{QA} = \frac{r_0}{q_0} \Rightarrow \frac{0.7}{0.4} = \frac{r_0}{3.7} : r_0 = 6.475$

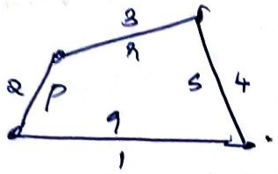
Same dir as Q

$\vec{V_S} = \vec{V_R} + \vec{V_{rs}}$.

Crank → Rotating link
Rocker → oscillating link
Slider → sliding link.

four bar chain.

Inversions and 4 bar mechanism:



p = shortest link.

if $\boxed{p+q \leq r+s}$

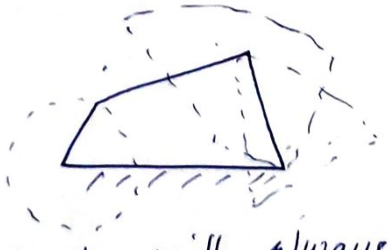
class I mechanism
or
grashof linkages

Grashof law:

$\boxed{p+q \leq \underbrace{r+s}}$ other 2 links.

Smallest link, largest link

Consider fixing link 1

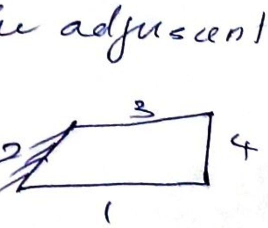


crank and Rocker lever mechanism

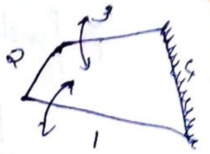
A crank will always be the shortest link
when we fix a link adjacent to a crank
we get crank rocker mechanism.

- when we fix the shortest link. both the adjacent links become crank (fully rotates

→ Double crank / Crank & crank Mechanism



(3) Fixing the link opposite to the crank, we get Double rocker mechanism



These 4 are class I chain Inversions $(p+q \le r+s)$

class 2 linkages

- if $p+q \ge r+s$

Which ever link you fix you will get

ie all inversions are Double rocker mechanism

class 3 linkages

$p+q = r+s$ (satisfy grashof law

A Inversion are similar to clas I linkages.
Except in some orientations, that these links become Collinear → Deltoid linkages